

Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

2-AB, 2-aminobenzamide; ALCOA+, attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available; AMV, analytical method validation; CHO, Chinese hamster ovary; CO₂, carbon dioxide; CPP, critical process

parameter; Cpk, one-sided process capability index; CQA, critical quality attribute; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; HPLC, high performance liquid chromatography; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; pCO₂, partial pressure of dissolved carbon dioxide; PRESS, prediction error sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SDM, scale-down model; SEC, size exclusion chromatography; SOP, standard operating procedure; UPLC, ultra performance liquid chromatography; WC-CPP, well-controlled critical process parameter.

1 Purpose and scope

The production bioreactor is the step at which the glycan, charge variant and aggregate quality attributes of A-Mab are formed. The purification train that follows carries those attributes through with little modification, and no step downstream of the harvest reforms a glycan or reverses a deamidation. For this reason the bioreactor is the step at which the design space for product quality is defined, and this plan sets out the studies that will define it.

The commercial process operates at 15,000 L of culture, a scale at which the number of runs a designed experiment needs cannot be supplied, and the studies will therefore be executed on a qualified scale-down model of that vessel. The plan states what will be studied, how the study will be run and how the results will be judged. The findings themselves belong to the paired report (PCR-003) and are not anticipated here.

The scope covers the 9 process parameters of the production bioreactor listed in Table 6.1, the 5 quality attributes that will be measured as responses, and the analytical methods that support them. The work is a Stage 1 process design activity in the sense of the lifecycle approach (U.S. Food and Drug Administration 2011), carried out under the enhanced development approach (International Council for Harmonisation 2009, 2012), and criticality is treated throughout as a continuum and not as a binary split into critical and non-critical attributes (International Council for Harmonisation 2023). Seed expansion up to the N-1 stage lies outside the scope, since prior platform knowledge covers it and PTP-001 records that position. The same scope excludes harvest and clarification, which PCP-004 covers.

Two limits apply to this plan. The first limit is that it does not qualify the scale-down model, which is qualified under SOP-1001 and reported separately. The second limit is that it does not by itself define the commercial control strategy, because host cell protein and residual DNA are formed in the bioreactor and cleared by later steps (PCP-005, PCP-007 and PCP-008).

2 Objectives

The study has five objectives, and each one produces a defined output in PCR-003 and a defined input to the control strategy the transfer will hand to the receiving site.

- (i) Quantify the effect of each parameter in Table 6.1 on the 5 quality attributes the production bioreactor forms, over ranges deliberately wider than the ranges the commercial process will use, and estimate the interactions between the parameters studied together.
- (ii) Define a multivariate operating region in the well-controlled culture parameters over which every attribute the step governs is predicted to meet acceptance, and locate the normal operating ranges of Table 6.1 inside that region.
- (iii) Establish a proven acceptable range for each parameter that governs an attribute, under both of the analyses described in §8, and state for each range the acceptance criterion it was judged against.
- (iv) Classify every parameter as a critical, well-controlled critical, key or general process parameter, using the decision logic of SOP-4001 together with the control capability the receiving site can demonstrate for that parameter.
- (v) Provide the ranges, the in-process controls and the monitoring commitments that the Stage 2 control strategy will carry for this step, with the evidence for each of them.

Objectives (i) to (iii) rest on the designed experiments in §6, while objective (iv) rests on those experiments together with the equipment capability the transfer confirms. Objective (v) is the output the transfer needs, and the first four objectives support it.

3 Related documents

The documents that govern this plan and the documents that consume the results are listed in Table 3.1.

Table 3.1: Documents related to this plan.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled documents the study will follow are listed in Table 3.2, where each modelled attribute has one validated analytical method and the validation report for that method holds the performance data.

Table 3.2: Controlled procedures and validated analytical methods.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

A-Mab is produced by fed-batch culture of a recombinant Chinese hamster ovary cell line in a stirred tank bioreactor. The culture is inoculated from the N-1 seed bioreactor, held at set-points for pH, temperature, dissolved oxygen and dissolved CO₂, and fed on a fixed schedule until the harvest criterion is met. Operation follows SOP-2003, and medium and feed are prepared under SOP-2004.

The platform is not new: three earlier humanized IgG1 products (X-Mab, Y-Mab and Z-Mab) have been manufactured on it at commercial scale, and A-Mab itself has been produced through clinical supply and through engineering runs. That experience establishes the mechanisms the study expects to see. The study therefore confirms and bounds behaviour at this step instead of discovering it. Several A-Mab acceptance limits were themselves set from X-Mab clinical experience on the same platform. Therefore, the platform is the source of some of the acceptance criteria as well as of the mechanistic expectations.

Three of the five modelled attributes are enzymatic in origin: galactosylation, high mannose and afucosylation. Galactosylation follows the activity of the Golgi galactosyltransferase and the supply of the nucleotide sugar donor, both of which decline as a culture ages and the intracellular environment deteriorates. High mannose reports how completely mannosidase trimming has run, which makes the attribute sensitive to luminal pH and to temperature (the two parameters that set enzyme activity most directly). Afucosylation is the one attribute of the three set by a competition, between core fucosyltransferase activity and the rate at which antibody leaves the Golgi, and the level is expected to fall as a culture is extended.

The other two modelled attributes are chemical in origin, and both are expected to depend on culture duration more than on any other parameter. Acidic charge variants arise mainly from deamidation of product already secreted into the broth, which proceeds faster at higher pH and accumulates over a longer culture. Aggregate has the same character, because antibody that sits longer in an ageing broth aggregates more. Both expectations point at culture duration as the parameter to bound. They are tested and quantified by the designed experiments in §6.

4.2 Quality attributes in scope

The quality attributes the production bioreactor sets are listed in Table 4.1 with the acceptance criterion and the criticality of each, and the score in the last column is the

Tool #1 impact and uncertainty score of the A-Mab risk methodology (CMC Biotech Working Group 2009).

Table 4.1: Quality attributes set by the production bioreactor.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2-13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10-40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3-10 % of glycans	VH	80
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

The highest score of the seven attributes in Table 4.1 belongs to high mannose, at 80, which is also the only one of very high criticality. Afucosylation and galactosylation follow, both of high criticality and both linked to the effector function that carries the mechanism of action. The same criticality applies to aggregates, which are the one attribute in the group that a later step reduces (PCP-007). Acidic charge variants sit at the other end of the register at 4, of very low criticality, and enter the study because the culture conditions move them and not because the drug substance is at risk from them.

Host cell protein and residual DNA are the two attributes formed in the bioreactor that are not modelled here, although both will be measured in the harvest of every run, because the load the purification train receives is an input to the clearance claims in PCR-005, PCR-007 and PCR-008. The 5 attributes that are modelled are the ones this step both forms and hands on unchanged.

4.3 Risk-based prioritization of parameters

RA-001 set the scope of this study, scoring each parameter on the potential impact of variation on a quality attribute and on the potential for interaction with other parameters, and deriving a study type from the combined score (CMC Biotech Working Group 2009). Of the 9 parameters in Table 6.1, 5 were assigned to the multivariate design and 4 to univariate assessment, and this plan executes that assignment without revisiting it.

The 5 multivariate parameters are culture pH, culture temperature, dissolved CO₂, osmolality and culture duration, and each of them has a mechanism on at least one glycan or charge attribute together with a plausible interaction with at least one of the others. Culture pH and culture temperature act on the same enzymatic steps, which is the clearest interaction the design has to resolve. Culture duration multiplies the exposure of every attribute to the conditions the other four parameters set, which makes an interaction with duration plausible in each case.

The 4 univariate parameters are dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume, and these four act on cell growth and on process performance, where platform experience across the three earlier products shows the effect on product quality to be small over the ranges proposed. A univariate assessment is adequate where a parameter has no plausible interaction with the multivariate set, and RA-001 found none for these four.

The ranges in Table 6.1 are the knowledge space and not the operating ranges, and each characterization range spans 1.6 to 2.5 times the width of the matching normal operating range. The extra width serves two purposes. It puts real distance between the design edges and the range the plant will use, which is what turns an effect estimate into a bound on operation. In addition, it leaves room for later process improvement inside a region that has already been studied (CMC Biotech Working Group 2009). No range extends into conditions under which the culture is known to fail.

5 Materials and methods

5.1 Scale-down model and its qualification

Execution will use bench-scale stirred tank bioreactors of equivalent design to the commercial vessels, operated under SOP-2003. The scale-independent parameters (pH, temperature, dissolved oxygen, initial viable cell concentration and culture duration) will be set to the values the commercial process uses. The scale-dependent parameters (agitation, gas flow rates, headspace pressure and working volume) will be set so that the bench system matches commercial process performance. Matching the commercial hardware settings themselves would not reproduce the culture environment, because the relation between an agitation rate and the shear a cell experiences changes with vessel geometry.

Qualification of the model is a precondition of execution and is not part of this study, and SOP-1001 governs it. Execution will not begin until the qualification report is approved. The qualification compares the bench system with 15,000 L data on the inputs and the outputs that matter here: the growth and viability profile, the titer and the 5 modelled attributes. A model that fails qualification for one attribute may still be used for the others, provided PCR-003 states the restriction.

The centre-point replicates in each design serve two purposes. They provide the pure error term against which lack of fit is tested (§5.5), and they show whether the bench system reproduces the set-point condition across the campaign. 3 centre points are included in the screening design and 4 in the response-surface design.

5.2 Cell line, media and culture operation

The cell line is the recombinant Chinese hamster ovary line used for A-Mab clinical supply, drawn from the qualified working cell bank, and basal medium and feed are chemically defined and prepared under SOP-2004. One lot of medium and one lot of feed will serve the whole campaign, which keeps lot variation out of the designed experiments.

Each run will be inoculated at the initial viable cell concentration the design specifies, controlled to the pH, temperature, dissolved oxygen and dissolved CO₂ set-points of that run, and fed on the platform schedule until the harvest day the design specifies. Daily samples support process monitoring, and the harvest sample supports the quality attributes.

Inputs held to platform specification are identity tested and quality controlled, and they are not characterized in this study: medium and feed lots, the gas supply and the working cell bank all fall in that group. A change to any input in that group changes the material basis of the study, and it would be assessed under change control before execution continued.

5.3 Analytical methods

Released N-glycan mapping by 2-AB HILIC-UPLC (SOP-3010, AMV-3010) reports afucosylation, galactosylation and high mannose from one chromatogram, size variants by SEC-HPLC (SOP-3011, AMV-3011) report aggregates, and charge variants by imaged capillary isoelectric focusing (SOP-3013, AMV-3013) report acidic variants. Host cell protein by ELISA (AMV-3012) and residual host cell DNA by qPCR (AMV-3014) characterize the harvest load.

Method performance is not restated in this plan. The performance figures (precision, accuracy and quantitation limits) are held in the validation reports named in Table 3.2, and PCR-003 will quote the relevant one wherever an effect approaches the precision of the method. Where scheduling allows, samples from all runs of one design will be assayed in a single analytical sequence, and every sequence carries a control sample, because drift between sequences would enter the model as run to run variation.

5.4 Sampling plan

One harvest sample per run is the primary sample of this study. It will be taken at the culture duration the run specifies, clarified by the standard bench procedure and split for the assays in §5.3. Sample handling, labelling and storage follow SOP-2003.

Daily in-process samples will be taken for viable cell density, viability and the main metabolites (for example glucose and lactate), and these support the interpretation of the harvest result and confirm that a run behaved as the design intended. They are not modelled.

In both designs, the centre-point runs carry the replicate structure. All of them are run at the set-points in Table 6.1, 3 of them in the screening design and 4 in the response-surface design. Centre points will be scheduled at the start of each execution block, in the middle and at the end, and a drift over the campaign will then appear as a difference between replicates.

5.5 Statistical methods

Analysis follows SOP-1002, and the factors are coded so that the set-point is zero and the edges of the characterization range are -1 and $+1$. The coded factors put every effect on one scale, which is the change in the response across half the studied range

(CMC Biotech Working Group 2009). Effects are reported in the natural units of the response.

The screening design will be analysed by ordinary least squares, with all main effects and all two-factor interactions in the model, and an effect is twice the coefficient on the coded factor, because one coded unit is half the characterization range. Every term in that model is assessed at $\alpha = 0.05$.

The response-surface design will be analysed with a full quadratic model in the retained factors, comprising the main effects, the two-factor interactions and the pure quadratic terms. The adequacy of that model will be judged on R^2 , on the adjusted and predicted forms of R^2 , on the overall F test and on an analysis of variance that partitions the residual into lack of fit and pure error. The predicted form is computed from PRESS residuals, which makes it a check on prediction and not on fit, while pure error comes from the centre-point replicates alone, which is why the replicate count matters more here than the total run count.

The two designs answer different questions: the screening design identifies which factors matter, and the response-surface design is the predictive model from which the operating region in PCR-003 will be derived. A screening model in 5 factors fitted to 19 runs is close to saturated, and a high fit statistic from such a model is not evidence of predictive quality.

For each attribute, capability will be estimated by Monte-Carlo simulation of 2,000 batches. Each simulated batch draws every parameter inside the normal operating range in Table 6.1, and the fitted response-surface model predicts each attribute for it. Capability is reported as a one-sided Cpk against the governing limit, with the margin to that limit. The Cpk estimate is conditioned on the scale-down model and on the fitted model. It is not a commercial-scale measurement, and Stage 2 provides that confirmation (U.S. Food and Drug Administration 2011).

6 Study design

6.1 Factors, ranges and study type

The parameters, the set-points, the ranges to be studied, the normal operating ranges and the study types are given in Table 6.1.

Table 6.1: Parameters to be studied, with the range studied and the normal operating range.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6-7.1	6.75-6.95	multivariate
Culture temperature	°C	35	34-36	34.5-35.5	multivariate
Dissolved CO2 (pCO2)	mmHg	100	40-160	70-130	multivariate
Osmolality	mOsm	400	360-440	375-425	multivariate
Culture duration	days	17	15-19	16-18	multivariate
Dissolved oxygen	% sat	50	30-70	40-60	univariate
Initial viable cell conc.	e6 cells/mL	1	0.5-1.5	0.8-1.2	univariate
Basal medium concentration	X	1.2	0.8-1.6	1-1.4	univariate
Nutrient feed-1 volume	% WV	12	9-15	10-14	univariate

For the design matrices in Appendix A and Appendix B, the letter codes are given in Table 6.2.

Table 6.2: Factor legend for the coded design matrices.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO2 (pCO2)	mmHg	screening + RSM
D	Osmolality	mOsm	screening

Code	Factor	Unit	Studied in
E	Culture duration	days	screening + RSM

Culture pH will be studied over 6.6–7.1, against a normal operating range of 6.75–6.95, and that width covers the excursions a commercial vessel can show during a base addition into a large working volume. Culture temperature will be studied over 34–36 °C, which is wider than the control system is capable of drifting. Osmolality will be studied over 360–440 mOsm, which covers the rise a full feed schedule produces. Dissolved CO₂ will be studied over 40–160 mmHg, because CO₂ accumulates at large scale and a commercial vessel is expected to sit higher than a bench vessel held at the same set-point (CMC Biotech Working Group 2009). Culture duration will be studied over 15–19 days, which brackets the platform harvest day on both sides. The 4 univariate parameters will be studied over the ranges in the same table, one at a time (§6.4).

6.2 Screening design

The screening design is a two-level fractional factorial in the 5 factors, with 16 factorial runs and 3 centre points, 19 runs in all, against the 32 runs a full factorial in those factors would need. The fractional design is of resolution V. Therefore, the design estimates every main effect clear of every two-factor interaction, and it aliases each two-factor interaction only with a three-factor interaction. The planned matrix is given in Appendix A.

At the screening stage, three-factor and higher interactions are assumed to be negligible, an assumption that is standard at that stage and that platform experience gives no reason to doubt for this step. However, a fractional design of this kind cannot resolve every interaction, and the factors it retains go forward into the response-surface stage for that reason.

The screening stage identifies. It does not define an operating region, and PCR-003 will not present a screening fit as a predictive model.

6.3 Response-surface design

The response-surface design is a face-centred central composite in the 4 factors that screening is expected to retain: culture pH, culture temperature, culture duration and dissolved CO₂. It comprises 16 factorial runs, 8 axial runs and 4 centre points, 28 runs in all. The axial runs sit on the faces of the cube, at the edges of the characterization ranges, and a rotatable design would place them outside the ranges Table 6.1 defines. Consequently, no run in this study leaves the knowledge space.

The second stage exists for curvature, because a two-level design cannot separate a curved response from a linear one and the centre points of the screening design can show only that curvature is present. The planned matrix is given in Appendix B.

Although the matrix in Appendix B assumes that screening retains the four factors named above, screening may retain a different set, and the matrix will then be re-issued before execution. A re-issue on that ground is an amendment to this plan and not a deviation from it.

6.4 Univariate assessment

The 4 univariate parameters are dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed-1 volume, and each of the four will be assessed separately, at the low edge of the characterization range, at the set-point and at the high edge. While one parameter moves, the other three and all 5 multivariate factors are held at the set-points in Table [6.1](#).

By construction, a univariate assessment cannot detect an interaction between the parameter it moves and the parameters it holds, and that limitation is accepted here on the basis of RA-001, which found no plausible interaction between these four parameters and the multivariate set. However, should a univariate result contradict that finding, the parameter will be carried into a multivariate follow-up study and PCR-003 will record the decision.

7 Acceptance and decision criteria

The criteria below are fixed before execution, because a criterion agreed after the results are known describes them instead of testing them.

The scale-down model is acceptable when the qualification under SOP-1001 shows no statistically significant difference between scales in the growth profile, the titer and the 5 modelled attributes. Execution of the study begins once that qualification report is approved.

A factor is carried from screening into the response-surface stage when a main effect, or an interaction containing it, reaches significance at $\alpha = 0.05$ for at least one attribute. A factor may also be carried on mechanistic grounds, where the estimated effect is of the same order as the significant terms and the residual degrees of freedom are few, and any factor carried on that second ground will be identified as such in PCR-003.

A response-surface model is adequate for the operating region when the overall regression is significant at $\alpha = 0.05$, when the adjusted and predicted forms of R^2 are consistent with each other and when the lack-of-fit test is not significant against pure error. However, a model that fails the lack-of-fit test may still support a restricted region, and PCR-003 will state the restriction and the reason for it. Where a lack of fit cannot be resolved by reducing the model or by restricting the region, the study has failed for that attribute, and the attribute is then held by monitoring and by release testing until further work is completed.

The operating region is acceptable when every attribute the step governs is predicted to meet the criterion in §8 throughout the region, including at the corner where the prediction is least favourable. The normal operating ranges in Table 6.1 must lie inside the region with margin, and if they do not, the operating ranges will be narrowed and the criterion will not be relaxed to accommodate them.

At the end of the study, every parameter will be given a class under SOP-4001: critical process parameter, well-controlled critical process parameter, key process parameter or general process parameter. The class follows from the demonstrated effect on a quality attribute and from the risk that the parameter leaves the region during commercial operation (CMC Biotech Working Group 2009). A parameter with no significant effect is still classified, and the absence of an effect is kept in the knowledge space as evidence of robustness over the ranges studied.

No numeric capability threshold is set in this plan. Capability will be reported for each attribute as a one-sided Cpk with the margin to the limit, and the acceptability of the set is judged in PCMR-001 against the framework in PCMP-001. A capability estimated from a scale-down model is a prediction, and confirmation at commercial scale is a Stage 2 activity (U.S. Food and Drug Administration 2011).

Any departure from this plan during execution will be recorded, investigated and impact assessed, and PCR-003 will carry the assessment. A deviation that invalidates a run requires that run to be repeated before the affected design is analysed.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of one parameter over which a quality attribute stays within acceptance while the other parameters are held or varied in a stated way. The range is univariate and the design space is multivariate. The design space and the ranges are kept apart in PCR-003, and this section states in advance how each range is derived.

The criterion a range is judged against comes first. For each of the 5 modelled attributes the criterion is the drug-substance specification in Table 4.1, unless a tighter in-process limit applies to the bioreactor harvest itself. Two kinds of in-process limit apply at this step. For the three glycan attributes and for acidic variants, none of which the purification train clears or modifies, the limit will be set from the capability this study demonstrates, at 2.5 standard deviations from the mean. For aggregates the limit is carried back from the drug-substance specification through the clearance the downstream steps deliver, then divided by an assurance margin of 6.8. The analysis reads the applicable criterion for each attribute from the register, and no criterion is calculated by hand.

For every pairing of an attribute with a response-surface factor, two analyses will be run. The first holds the other factors at the set-points in Table 6.1 and scans the parameter of interest across the full characterization range, evaluating the fitted response-surface model at 201 grid points. The second propagates normal operating variation: at each of 81 grid points the other factors are drawn within the normal operating ranges in Table 6.1, 2,000 draws at each point, with the residual variance of the model added to every prediction.

The decision rule differs between the two analyses. Under the first, the proven acceptable range is the contiguous interval containing the set-point over which the predicted mean stays within acceptance. Under the second, the interval must hold the whole 95 % predictive interval within acceptance. The second rule is a robustness criterion and will give the narrower answer wherever the other factors have an effect. Both ranges will be reported for every pairing, in one table.

Each governed attribute will also be presented as a figure against the parameter that governs it, in two panels that correspond to the two analyses, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked. Where a proven acceptable range covers the whole characterization range, the parameter does not constrain the step over the region studied. The binding constraint on the step is then the multivariate corner of the design space, and not any univariate range.

Every proven acceptable range this study reports will carry three bounds. Nothing is proven outside the characterization range, because nothing outside it was studied. The range is a property of the fitted response-surface model, and a model that fails the adequacy criteria in §7 supports no proven acceptable range at all. Finally, the second analysis propagates the normal operating variation of the other response-surface factors and the residual variance of the model, and it does not propagate variation in the parameters assessed univariately.

9 Data management and integrity

Throughout the study, records will be attributable, legible, contemporaneous, original and accurate, and they will also be complete, consistent, enduring and available (ALCOA+).

Raw analytical data are acquired and retained in the validated chromatography and electrophoresis data systems, with audit trails enabled, and bioreactor process data are acquired by the control system and retained in the process historian. The design matrices, the calculated results and the statistical analyses are held in a version-controlled study workbook, from which the analysis will be reproducible using the raw data and the recorded model settings.

Before analysis, every transcription of a result will be verified by a second person. The statistical analysis will be reviewed by someone other than the analyst who performed it. Records are retained under the site record retention procedure.

10 Roles and responsibilities

Process Development and MSAT own this plan. Process Development and MSAT also design the study, execute the bioreactor runs, perform the statistical analysis and write PCR-003.

Analytical Development performs the assays in §5.3, maintains the validated methods and reports the analytical results. Manufacturing Science reviews the design and the proposed ranges against the capability of the commercial equipment, and confirms that the ranges the study proposes can be held at the receiving site. Statistics reviews the design, the model forms and the analysis. Quality Assurance approves this plan and PCR-003, and disposes any deviation raised during execution.

11 Deliverables and schedule

The deliverable is PCR-003, the process characterization report for the production bioreactor, which carries the effects, the models, the operating region, the proven acceptable ranges, the capability estimates and the parameter classifications, and which rolls up into PCMR-001.

The work runs in four phases, of which qualification of the scale-down model under SOP-1001 comes first, followed by execution and analysis of the screening design and then by execution and analysis of the response-surface design. Once the analytical results for every run of a phase are available, analysis of that phase begins. The univariate assessments may run in parallel with either designed experiment. Because the campaign schedule is held in PTP-001, no calendar dates are given here.

12 Risks and assumptions

The principal risk to the study is that the scale-down model does not represent the commercial vessel for the attributes being modelled, and dissolved CO₂ is where that risk concentrates, because CO₂ accumulation depends on scale and a bench vessel strips it more readily than a commercial one (CMC Biotech Working Group 2009). Including dissolved CO₂ as a factor over 40–160 mmHg is the mitigation, since the range covers the levels a commercial vessel reaches whether or not the bench vessel reaches them by the same route.

Material supply is the second risk. Using one lot of medium and one lot of feed across the campaign keeps lot variation out of the model, and it also means the study says nothing about lot variation. That variation is handled by the incoming material specification and by SOP-2004, and not by this study.

Assay variability is the third risk. Where an effect is small relative to the precision of the method, PCR-003 will attribute the variability to the assay and will hedge the conclusion accordingly. That variability is estimated from the centre-point replicates, which give the combined process and assay term such a judgement needs.

Three assumptions are open to challenge and are stated here for that reason. Three-factor and higher interactions are assumed negligible in the screening design. The model fitted to the response-surface design is assumed to hold across the whole characterization region, which the lack-of-fit test examines and cannot prove. Finally, the platform mechanisms in §4.1 are assumed to apply to A-Mab, and the study confirms or refutes them for each of the five modelled attributes.

13 Approvals

This plan takes effect once the functions listed in Table 1 have approved it. Execution begins after that approval and after the scale-down model qualification report is approved. Any change before execution is complete will be issued as a revision of this plan and approved by the same functions.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.

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U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned screening design matrix

The planned screening matrix is given in Table 13.1, in coded form. The letter codes are those of Table 6.2, and the coded levels -1 , 0 and $+1$ correspond to the low edge, the set-point and the high edge of each characterization range in Table 6.1. Response columns are absent because no run has been executed.

Table 13.1: Planned screening design matrix, coded.

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

Appendix B — Planned response-surface design matrix

The planned response-surface matrix is given in Table 13.2, in the same coded form. The factorial runs occupy the corners of the cube, the axial runs sit at the centre of each face and the centre points sit at the set-point condition.

Table 13.2: Planned response-surface design matrix, coded.

Run	Type	A	B	E	C
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0